Intermediate Microeconomics Practice 1

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Instructions:

- Answer all questions in English.
- Write your calculation process clearly for computational problems.
- For graphing questions, clearly label all axes, curves, and equilibrium points.

Part I: True or False (20 points)

Indicate whether the following statements are True (T) or False (F). Briefly explain your reasoning for any False statements.

1. **[ ]** Walras' Law states that if $n - 1$ markets in an exchange economy are in equilibrium, the n-th market must also be in equilibrium.
2. **[ ]** In a First-Degree (Perfect) Price Discrimination scenario, the monopolist extracts all consumer surplus, resulting in zero deadweight loss.
3. **[ ]** A pure public good is characterized by being rival in consumption and excludable.
4. **[ ]** In the Prisoner's Dilemma game, the Nash Equilibrium is always Pareto efficient.
5. **[ ]** Adverse selection usually occurs *before* the transaction takes place due to hidden information, whereas moral hazard occurs *after* the transaction due to hidden action.
6. **[ ]** Under Third-Degree Price Discrimination, a monopolist will charge a higher price in the market with more elastic demand (more negative price elasticity).
7. **[ ]** According to the Coase Theorem, if property rights are well-defined and transaction costs are zero, the initial allocation of property rights determines whether the efficient outcome can be achieved.
8. **[ ]** The "Core" of an exchange economy is the set of all Pareto-optimal allocations that are welfare-improving for both consumers relative to their initial endowments.
9. **[ ]** In a natural monopoly, setting the price equal to Marginal Cost ($P = MC$) allows the firm to break even (zero economic profit).
10. **[ ]** In a mixed-strategy Nash Equilibrium, a player chooses probabilities such that the opponent is indifferent between their pure strategies.

Part II: Multiple Choice (20 points)

Select the best answer for each question.

1. A monopolist faces a demand curve with constant elasticity $\varepsilon = -3$. If the marginal cost of production is constant at \$20, what is the profit-maximizing price?

   A. \$20

   B. \$30

   C. \$40

   D. \$60

2. Which of the following best describes the ”Tragedy of the Commons”?

   A. A failure of public goods provision due to free-riding.

   B. Over-consumption of a non-excludable but rival resource.

   C. Under-production of a good with positive externalities.

   D. A coordination failure in a Stag Hunt game.

3. In an Edgeworth Box, the Contract Curve consists of:

   A. All allocations where one consumer is better off than the other.

   B. All allocations where the marginal rate of substitution (MRS) is equal for both consumers.

   C. All allocations inside the ”lens” formed by the initial indifference curves.

   D. Only the competitive equilibrium points.

4. Consider the ”Market for Lemons” (Adverse Selection). If buyers cannot distinguish between high-quality and low-quality cars, and only know the average value:

   A. Only high-quality cars will be traded.

   B. Both types will be traded at the average price.

   C. High-quality car sellers may exit the market, leading to a market collapse.

   D. Signaling has no effect on this market.

5. The efficiency condition for the provision of a public good with two consumers (A and B) is given by:

   A. $MRS_A = MRS_B = MRT$

   B. $MRS_A + MRS_B = MC$ (or MRT)

   C. $MU_A = MU_B$

   D. $MRS_A/P_A = MRS_B/P_B$

6. In a Sequential Game where Firm A moves first (Stackelberg leader) and Firm B moves second (follower):

   A. Firm A usually achieves a higher profit than in a simultaneous Cournot game.

B. Firm B has the "first-mover advantage."
C. The outcome is identical to the Cournot Nash Equilibrium.
D. Both firms produce the monopoly quantity.

7. If a monopolist implements a Two-Part Tariff ($T = F + p \cdot q$) to maximize total profit and extract all surplus from identical consumers, it should set the per-unit price p equal to:

   A. The monopoly price.
   B. Zero.
   C. Marginal Cost.
   D. Average Total Cost.

8. High-ability workers getting a college degree to distinguish themselves from low-ability workers, even if the degree adds no productivity, is an example of:

   A. Moral Hazard
   B. Screening
   C. Signaling
   D. Bundling

9. In the presence of a negative production externality (e.g., pollution), the private market equilibrium quantity is usually ______ than the socially optimal quantity.

   A. Lower
   B. Higher
   C. The same
   D. Unrelated

10. A situation where a consumer's utility depends on the consumption of others, or a production function depends on the activities of others, without market compensation, is defined as:

    A. An Externality
    B. A Public Good
    C. Imperfect Competition
    D. Asymmetric Information

Part III: Term Definitions (15 points)

Briefly explain the following terms in the context of microeconomics (2-3 sentences each).

1. **Pareto Efficiency**
2. **Nash Equilibrium**
3. **Moral Hazard**
4. **Third-Degree Price Discrimination**
5. **Free-Rider Problem**

Part IV: Calculations (30 points)

Show all steps of your work.

Question 1: Monopoly and Price Discrimination (10 points)

A monopolist sells a product in two separate markets, Market 1 and Market 2. The demand curves are:

$$P_1 = 100 - Q_1$$

$$P_2 = 80 - 2Q_2$$

The monopolist's total cost function is $TC(Q) = 20Q$, where $Q = Q_1 + Q_2$.

(a) If the monopolist can practice third-degree price discrimination, calculate the profit-maximizing quantity (Q_1, Q_2) and price (P_1, P_2) for each market.

(b) Calculate the total profit under price discrimination.

(c) Which market has a more elastic demand at the equilibrium point? Verify using the markup rule or elasticity formula.

Question 2: Game Theory and Mixed Strategies (10 points)

Consider the following payoff matrix for a game between Player A and Player B:

	B: Left	**B: Right**
A: Up	2, 3	0, 1
A: Down	1, 0	3, 2

(a) Does this game have any pure-strategy Nash Equilibria? If yes, identify them. If no, explain why.

(b) Find the mixed-strategy Nash Equilibrium. Let p be the probability Player A plays Up, and q be the probability Player B plays Left. Calculate the equilibrium values for p and q.

Question 3: Externalities (10 points)

A steel factory (Firm S) is located upstream from a fishery (Firm F). The steel firm's profit function is $\pi_S = 12s - s^2 - (x - 4)^2$, where s is steel output and x is pollution. The fishery's profit function is $\pi_F = 10f - f^2 - xf$, where f is fish output. The pollution x negatively affects the fishery.

(a) If the firms operate independently (steel firm ignores the external cost), what level of pollution x will the steel firm choose?

(b) If the two firms merge and maximize joint profit, what is the socially optimal level of pollution x^*?

(c) Comparing (a) and (b), does the independent market produce too much or too little pollution?

Part V: Graphing and Analysis (15 points)

Topic: General Equilibrium and the Edgeworth Box

Consider an exchange economy with two consumers, A and B, and two goods, 1 and 2. Consumer A has an initial endowment $\omega_A = (8, 2)$ and Consumer B has an initial endowment $\omega_B = (2, 8)$. Their preferences are standard (convex indifference curves).

(a) Draw an **Edgeworth Box** diagram. Clearly label the following:

- The dimensions of the box (total quantity of Good 1 and Good 2).
- The endowment point W.
- An indifference curve for A and an indifference curve for B passing through the endowment point.

(b) On your diagram, shade the area representing the **Lens of Trade** (the set of allocations that make both consumers better off).

(c) Illustrate and label the **Contract Curve**. Explain what condition holds true at every point on the Contract Curve.

(d) Identify the **Core** on the Contract Curve.

(e) Draw a budget line and a potential competitive equilibrium point. Explain how the price ratio (P_1/P_2) relates to the indifference curves at this equilibrium.